

Kevin Jose Real Estate Project (Frontend)

1. Introduction & Overview

The RealEstate project is a modern web application built with React.js to provide a seamless platform for real estate listings, bookings, and management. It includes an admin dashboard, property listing, authentication, and various UI enhancements for an optimal user experience.

2. Project Structure

The project follows a well-structured architecture to ensure maintainability and scalability. Below is the directory structure:

src/

```
| — Components/      # Common reusable components
| — Images/          # Static images
| — admin/           # Admin dashboard components
|   | — Componets/    # Admin UI components
|   | — Image/        # Image management
|   | — Navbar/       # Admin navigation
|   | — Pages/        # Admin pages
|   | — Style/        # SCSS and styling
|   | — redux/        # Redux store and slices
| — routes/          # Route protection and navigation
|   | — ProtectedRoute.js # Wrapper for protected admin routes
|   | — privateRoute.js  # General private route handling
| — services/user/    # API service functions for users
| — App.js            # Main application entry point
| — index.js          # Root rendering file
| — App.css           # Global styles
| — .gitignore        # Git ignore file
```

3. Installation & Setup Guide

Prerequisites

- Node.js (v16 or later)

- npm or yarn

Installation Steps

Clone the repository

git clone <https://github.com/Spangles-Infotech/Kevin-Jose.git>

cd realestate

Install dependencies

npm install

Start the development server

npm start

The application will run at <http://localhost:3000/> by default.

4. Technology Stack & Dependencies

Core Technologies

- React.js (Frontend Framework)
- Redux Toolkit (State Management)
- React Router v6 (Routing)
- Axios (API Calls)
- SCSS & Bootstrap (Styling & UI Components)

Key Dependencies

```
"dependencies": {  
  "@reduxjs/toolkit": "^2.2.5",  
  "axios": "^1.6.7",  
  "bootstrap": "^5.3.2",  
  "react": "^18.2.0",  
  "react-bootstrap": "^2.9.1",  
  "react-calendar": "^4.8.0",  
  "react-hook-form": "^7.51.5",
```

```
"react-icons": "^4.12.0",  
"react-router-dom": "^6.23.0",  
"react-toastify": "^10.0.4"  
}
```

5. Features & Functionality

User Features

- ✓ View property listings with filters and search functionality
- ✓ Book properties through an interactive UI
- ✓ Login/Signup with OTP verification
- ✓ Responsive design for mobile and desktop

Admin Features

- ✓ Secure login with Protected Routes
- ✓ Add, edit, delete property listings
- ✓ View and manage user bookings
- ✓ Dashboard analytics and reports

6. Routing & State Management

Routing (React Router v6)

```
import { BrowserRouter as Router, Routes, Route } from "react-router-dom";  
  
import Home from "../Pages/Home";  
  
import AdminDashboard from "../admin/Pages/Dashboard";  
  
import ProtectedRoute from "../routes/ProtectedRoute";  
  
function App() {  
  return (  
    <Router>  
      <Routes>  
        <Route path="/" element={<Home />} />  
        <Route path="/admin" element={(  
          <ProtectedRoute>  
            <AdminDashboard />  
          </ProtectedRoute>  
        )} />  
      </Routes>  
    </Router>  
  );  
}
```

```

    </Routes>
  </Router>
);
}
export default App;

```

State Management (Redux Toolkit)

```

import { configureStore } from '@reduxjs/toolkit';
import userReducer from './redux/userSlice';

export const store = configureStore({
  reducer: {
    user: userReducer,
  },
});

```

7. API Integration & Services

Example API Call with Axios

```

import axios from 'axios';

export const fetchProperties = async () => {
  try {
    const response = await axios.get('https://api.example.com/properties');
    return response.data;
  } catch (error) {
    console.error('Error fetching properties', error);
    throw error;
  }
};

```

8. How to Contribute & Extend

Contributing

1. Fork the repository.

2. Create a new branch (feature/new-feature).
3. Commit your changes with descriptive messages.
4. Push your changes and create a Pull Request.

Extending the Project

- Add a property comparison feature for users.
- Implement Google Maps integration for better location display.
- Develop real-time chat for buyers and sellers.

9. Conclusion

The Kevin Jose Real Estate Project provides a scalable, feature-rich real estate platform built with React.js. With strong architecture and best practices, the project can be extended for future needs.